

Rule based engine

	Metric (Data point)	Rule/ Condition	Alert level	Alert message/ action
1	HbA1c	• >7% for those without complication (IHD/ HF, CVA/Stroke, nephropathy) • >7.5% for those with complications (Age 70-<75, on insulin, duration of diabetes is >30?) • >8% for those with complications (Age >75-80, on insulin, duration of diabetes is >30?)	High	HCP Alert: Patient [Patient Name] HbA1c >7%. Review care plan for glycemic control. Patient Alert: Your HbA1c is elevated. Please consult your doctor for review.
2	BP	• >130/80 mmHg for patients aged <65years • >140/90 mmHg for Patients aged >65years • <90/60mmHg +/- symptoms (dizziness, unexplained tiredness)	High High Critical	Similar kind
3	LDL	• >70mg/dL for those with high risk for CVD • >100mg/dL for those with low risk of CVD	High Moderate	HCP Alert: Patient [Patient Name] (high CVD risk) LDL >70. Review lipid-lowering therapy. Patient Alert: Your LDL cholesterol is high. Discuss with your doctor.
4	ASCVD risk score (10-year) Patient meets criteria for calculation (e.g., age 40-79 years, no	• Low risk < 7.5%	Moderate	HCP Alert: Patient [Patient Name] has a newly calculated 10-year ASCVD risk score of [Score]%. Review for primary prevention strategies.

	prior ASCVD, diabetes, or severe hypercholesterolemia)	Intermediate risk >7.5% to <20%	High	Patient Alert: Your doctor has calculated your 10-year risk of heart disease or stroke. They will discuss this with you at your next visit. HCP Alert: Patient [Patient Name] 10-year ASCVD risk is [Score]% (Intermediate Risk). Strongly consider statin therapy and intensive lifestyle modification counseling. Patient Alert: Your risk of heart disease or stroke in the next 10 years is elevated. Your doctor will discuss lifestyle changes and possibly medication to help lower this risk. HCP Alert: Patient [Patient Name] 10-year ASCVD risk is [Score]% (High Risk). Initiate/intensify statin therapy and aggressive lifestyle modification immediately. Patient Alert: Your risk of heart disease or stroke in the next 10 years is high. It's very important to follow your doctor's recommendations for lifestyle changes and medication to significantly lower this risk.
		High risk >20%	Critical	

		• Significant Increase in Risk: ASCVD risk score increases by > 5 percentage points from previous calculation within a 1-year period (e.g., from 8% to 13%). • Overdue for Re-calculation: Last ASCVD risk calculation was more than 1 year ago (or as per guideline frequency, e.g., every 4-6 years for low-risk individuals).	High Moderate	HCP Alert: Patient [Patient Name] ASCVD risk has significantly increased to [New Score]% from [Old Score]%. Re-evaluate risk factors and intensify interventions. Patient Alert: There has been a notable change in your heart disease and stroke risk assessment. Your doctor will discuss this with you. HCP Alert: Patient [Patient Name] ASCVD risk score is overdue for re-calculation. Update relevant parameters and assess current risk. Patient Alert: It's time to re-evaluate your risk of heart disease and stroke. Your doctor will discuss this with you.
5	Urine ACR	• ACR >30- 300 mg/g (micro/macroalbuminuria) •	High	Similar kind HCP Alert: Patient [Patient Name] showing micro/macroalbuminuria. Repeat Urine ACR test. On repeat test if still high, to review for medication optimization (ACEi/ ARBs/ SGLT2i/ GLP-1 analogue/ MRA) . Patient Alert: Your kidney

				test shows significant changes. Repeat Urine ACR test once more and if still report shows high value/ changes, it may require medication review and medication changes
6	Serum creatinine	- Significant increase from baseline (>25% increase or rise >0.5 mg/dL) - eGFR <60	High	HCP Alert: Patient [Patient Name] creatinine elevated. Assess kidney function/medication. Patient Alert: Your kidney function test is elevated. Please consult your doctor.
7	Serum potassium	<3.5mEq/L (Hypokalemia) >5mEq/L (Hyperkalemia)	High High	
8	PHQ 2	Score >3	Moderate	HCP Alert: Patient [Patient Name] PHQ-2 score suggests depressive symptoms. Consider PHQ-9. Patient Alert: Your mood questionnaire suggests you might need support. Talk to your doctor.
9	GAD 2	Score >3	Moderate	Similar kind
10	PHQ 9	- Score >=10 (Moderate depression) - Score >=20 (Severe depression)	High Critical	Similar kind HCP Alert: Patient [Patient Name] PHQ-9 score indicates severe depression. Urgent psychiatric assessment. Patient Alert: Your mood is very low. Seek immediate medical

				help.
11	GAD 7	- Score ≥ 10 (Moderate anxiety) - Score ≥ 20 (Severe anxiety)	High Critical	Similar kind
12	Kessler's scale (K-10)	- Score 16-21 (Moderate) - Score 22-29 (High) - Score 30-40 (Very high)	High	HCP Alert: Patient [Patient Name] Kessler score indicates distress. Further psychological evaluation. Patient Alert: Your stress levels are high. Talk to your doctor for support
13	Stop BANG score for OSA	- Score 3-4 (Intermediate risk) - Score 5-8 (High risk)	Moderate Critical?	HCP Alert: Patient [Patient Name] high risk for OSA. Consider sleep study referral. Patient Alert: You may be at risk for sleep apnea. Discuss with your doctor.
14	Medication adherence monitoring recommendation Weekly recall Number of pills missed divided by number of pills advised- Percentage Number of pricks missed divided by number of pricks advised (for insulin)- Percentage	- Adherence below target (e.g., $<80\%$ based on self-report) - Critical non-adherence: Medication adherence $<50\%$ or missed dose of critical medication (e.g., insulin, anti-hypertensive) for >3 consecutive days (as detected by self-report) - Medication adherence $50-79\%$ for >7	Moderate? High	HCP Alert: Patient [Patient Name] medication adherence low. Review barriers/education. Patient Alert: Important: Your medication adherence seems low. Let's work together to improve this. HCP Alert: Patient [Patient Name] critical non-adherence to vital medications. Urgent follow-up. Patient Alert: Critical: You're consistently missing important medications. Please contact your doctor

		consecutive days or missed >2 doses of non-critical meds in a week • Reported Side Effects/Issues: Patient reports new or worsening side effects from medication or difficulty with administration	Moderate Moderate	immediately. HCP Alert: Patient [Patient Name] suboptimal medication adherence. Review barriers to adherence/education. Patient Alert: Important: Your medication adherence seems low. Let's work together to improve this for better health. HCP Alert: Patient [Patient Name] reports medication side effects/issues. Review/adjust meds. Patient Alert: You're experiencing medication side effects or difficulties. Please talk to your doctor about this.
15	Follow up visit consultation monitoring recommendation (This can be as per recommendation by doctor) - Internal medicine/ Endocrine - Cardio - Opthal - Neuro - Nephro	• 3-6 months (Internal medicine) for patients with well controlled HbA1c <7% • 2-4 weeks initially or even earlier until targets are met and stable for - newly diagnosed patients or existing patients with consistently elevated HbA1c >7/7.5/8 as per patient age or uncontrolled BP >160/100mmHg - initiation of new medication or insulin, dose change of existing medication or significant alteration in treatment regimen - New onset of	High	

		micro/macrovacular complications (e.g., microalbuminuria, new retinopathy, neuropathy symptoms, angina, kidney function decline) or worsening of existing complications. - Patient self-reports or platform detects critical symptoms: acute chest pain, hypoglycemia, acute deterioration in vision, severe neurological symptoms, unexplained rapid weight change • Weekly or bi-weekly digital check-ins, 1-2 months for in-person review) - Consistent non-adherence to medication, diet, or lifestyle recommendations • Weekly or bi-monthly: Elevated scores on mental health screening tools	Critical Critical High High	
16	Blood pressure monitoring recommendation	• BP >180/120mmHg with acute target organ damage symptoms (e.g., severe headache, chest pain, vision changes, shortness of breath) • BP >180/120 mmHg without above mentioned symptoms	Critical High	HCP Alert: Patient [Patient Name] BP Crisis with Symptoms. Urgent ER referral/Immediate consult. Patient Alert: Your blood pressure is extremely high with urgent symptoms. Seek immediate emergency medical attention. HCP Alert: Patient [Patient Name] BP

		• Consistently High BP (Grade 2/3): Average of > 3 readings over 1 week: >160/100mmHg	High	Urgency. Urgent clinic visit recommended within 24 hours. Patient Alert: Your blood pressure is very high. Contact your doctor or clinic immediately. HCP Alert: Patient [Patient Name] consistently high BP. Review treatment strategy urgently. Patient Alert: Your blood pressure is consistently high. Please consult your doctor for immediate review.
		• Consistently High BP (Grade 1): Average of > 3 readings over 1 week: 140-159/90-99 mmHg	Moderate	HCP Alert: Patient [Patient Name] consistently elevated BP. Consider medication adjustment/lifestyle. Patient Alert: Your blood pressure is consistently elevated. Please consult your doctor for review.
		• < 90/< 60 mmHg WITH symptoms (e.g., dizziness, fainting, weakness)	Critical	HCP Alert: Patient [Patient Name] symptomatic hypotension. Urgent assessment needed. Patient Alert: Your blood pressure is very low, and you have symptoms. Seek immediate medical attention.
		• < 90/< 60 mmHg without above mentioned symptoms	Moderate	HCP Alert: Patient [Patient Name] asymptomatic hypotension. Review

		• Missing BP Readings: No BP data uploaded for >3 consecutive days (configurable frequency based on patient plan) - Advised to upload - Not advised • High BP Variability: Significant fluctuations (e.g., >30 mmHg systolic difference within 24 hours, not explained by activity)	High Low Moderate	medications/hydration. Patient Alert: Your blood pressure is low. Please discuss this with your doctor HCP Alert: Patient [Patient Name] missing BP readings. Send reminder. Patient Alert: Remember to upload your blood pressure readings regularly as per your plan. HCP Alert: Patient [Patient Name] BP shows high variability. Investigate causes (meds, adherence, lifestyle). Patient Alert: Your blood pressure readings are fluctuating significantly. Please discuss this with your doctor.
17	Blood sugar monitoring recommendation • FBS	• FBS <54mg/dL (Severe hypoglycaemia) with severe symptoms requiring assistance or altered mental status • FBS <70mg/dL (Moderate hypoglycaemia) with symptoms • FBS > 200mg/dL (Severe hyperglycemia) consistently (e.g., >2 consecutive readings)	Critical High Critical	Similar kind Similar kind HCP Alert: Patient [Patient Name] consistent very high FBS. Urgent therapy

	- PPBS/GRBS - Missed blood sugar readings	- FBS > 130-200 mg/dL Hyperglycemia (Uncontrolled) consistently (e.g., average of >3 readings over 1 week) - PPBS > 220 mg/dL Hyperglycemia (Severe): consistently (e.g., >2 consecutive readings 2 hours after meal) - PPBS >180-220 mg/dL Hyperglycemia (Uncontrolled) consistently (e.g., average of >3 readings over 1 week 2 hours after meal) - PPBS <70 - PPBS 71-100 - No BG data uploaded for >24 hours (configurable frequency based on patient plan) - Advised by doctor - Not advised Patient plan????	High Critical High Critical Moderate High Low	adjustment. Patient Alert: Your fasting blood sugar is very high. Please contact your doctor for immediate review. HCP Alert: Patient [Patient Name] uncontrolled PPBS. Review mealtime diabetes management. Check for urine ketones Patient Alert: Your after-meal blood sugar is consistently high. Please consult your doctor. HCP Alert: Patient [Patient Name] missing blood sugar readings. Send reminder. Patient Alert: Remember to upload your blood sugar readings regularly as per your plan.
	Blood sugar Monitoring Recommendation	Poor Control: HbA1c is high as defined above +/- Insulin use +/- Non adherence to medications/ insulin FBS >200 AND/OR PPBS >220		Post consultation advice recommendation in HCP dashboard : daily pre meal and post meal monitoring atleast 3-4 readings per day for atleast 5-8

		• Suboptimal Control FBS>130-200 AND/OR PPBS > 180-220)	days. Recommend counseling on consistent measurement. Post consultation advice recommendation and instruction to patient : To help manage your diabetes effectively, please measure and upload your blood sugar <i>every day</i>. before meal and post meal monitoring at least 3-4 readings per day for at least 5-8 days, fasting/ before breakfast/ 2 hours after breakfast/ pre lunch/ 2 hours post lunch/ pre dinner/ 2 hours post dinner- fill in this blood sugar log tracker. Post consultation advice recommendation in HCP dashboard: At least 3-4 times monitoring per week at different time points. Post consultation advice recommendation and instruction to patient: Please ensure you are regularly measuring and uploading your blood sugar, for at least 3-4 times per week at different
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				time points such as fasting/ before breakfast/ 2 hours after breakfast/ pre lunch/ 2 hours post lunch/ pre dinner/ 2 hours post dinner- fill in this blood sugar log tracker.
	BP Monitoring Recommendation	• Poor BP Control >160/100mmHg +/- non adherence to antihypertensive drugs • Suboptimal BP Control >140-159/90-100 +/-non adherence to antihypertensive drugs		Post consultation advice recommendation in HCP dashboard: daily home BP monitoring at different time intervals. Emphasize proper technique and consistent logging. Post consultation advice recommendation and instruction to patient: To effectively manage your blood pressure, please measure and upload your blood pressure readings <i>every day recording at different time of the day.</i> Post consultation advice recommendation in HCP dashboard: increase BP monitoring, at least 3-4 times per week at different time interval). Post consultation advice recommendation and instruction to patient: Please ensure you are

				regularly measuring and uploading your blood pressure atleast, 3-4 times per week at different time of the day.
18	CV symptoms	• New onset Chest Pain, Shortness of Breath (acute), Palpitations (new/severe) • Discomfort in the chest/ upper back/both shoulders/neck/jaw or uneasiness/heaviness in your chest region or choking sensation? (sudden onset, increase on exertion, reduce on rest • Difficulty in breathing- worsen on lying down, better on sitting up, associated dry cough, cough more on lying down • Swelling of limbs or face or abdomen- Foot ware or chappals getting tight, feel bloating sensation, not eating well and always feel full • Fatigue- deviation from baseline physical activity, daily routine activities such as going to bathroom or taking bath, breathless or fatigue and feel like sitting or taking rest • Heart beating fast and sensation of skipping a heartbeat • Intermittent headaches, blurring vision and blood pressure at that time	Critical	HCP Alert: Patient [Patient Name] reported new/severe CV symptoms. ER referral/Urgent consult. Patient Alert: You have reported urgent heart-related symptoms. Seek immediate emergency medical help.
19	Nephro symptoms	• New onset significant edema, decreased urine output, flank pain	Critical	Similar kind

		(new/severe) • Frothy urine- persisting for two weeks or more (even after resolution of infection or menstruation if any), increase after strenuous work • Early morning Face swelling (started first) and leg swelling remaining more than a week • Significant amount of urine passing in the night associated with less amount of urine output in the day time • Shortness of breath when active or lying down- more tired or weak than normal, cough or wheeze, coughing up blood, nausea, lack of appetite, not able to exercise or be active as before • History of urine infection- 3 or more episodes per year or 2 or more episodes per 6 months, multiple antibiotic courses, any procedure or surgery for the same • Any blood in the urine • Any increased frequency or urgency or difficulty in passing urine or incomplete voiding		
20	Neuro symptoms	• New onset severe headache, sudden weakness/numbness, speech difficulty, vision changes, confusion • Transient weakness of hands or legs- sudden	Critical	Similar kind

		onset, towards one side, not able to lift hand or leg, inability to carry out daily activities (bathroom, taking bath, combing, dressing etc) • Tingling or numbness or burning sensation • Decreased touch and pain sensation • Reduced vision or transient blindness • Imbalance while walking • Not able to talk • Spinning of head or vomiting		
21	Retino symptoms	• Sudden vision loss, severe eye pain, floaters/flashes (new onset) • Reduced vision • Black spots in front of your eyes • Flashes of light or lightening in front of eyes • Abnormal shapes of known objects	High	Similar kind
22	Diet	• Consistent reporting of unhealthy diet/non-adherence to dietary plan for >7 days • Not adherent to high fibre diet or complex grains, portion size as food plate principle, 4-5 servings of fruits and vegetables, appropriate timing of meals and healthy snacks, no red meat, lean meat <2-3 times a week • Eating out or ordering food more than once a week	Moderate	HCP Alert: Patient [Patient Name] consistent diet non-adherence. Provide dietary counseling. Patient Alert: Let's review your diet plan. Consistent healthy eating is important.
23	Physical activity issues	• Consistent reporting of low/no physical activity	Moderate	HCP Alert: Patient [Patient Name] low

		- for >7 days (below personalized target) - Not adherent to strength training		physical activity. Encourage/recommend activity plan. Patient Alert: Regular activity is key. Let's find ways to increase your physical activity.
24	BMI	- BMI >23-25 Assess for - STOP BANG for OSA - FIB4 score for MASLD - BMI <18.5	High	HCP Alert: Patient [Patient Name] BMI >23. Discuss weight management strategy. Patient Alert: Your BMI is in the overweight to obese range. Talk to your doctor about weight management.
25	Sleep	Consistent reporting of poor sleep quality (>3 nights/week for >2 weeks) OR <5 hours sleep per night	Moderate	HCP Alert: Patient [Patient Name] reports persistent sleep issues. Investigate causes/refer. Patient Alert: Consistent poor sleep can affect your health. Talk to your doctor.
26	FIB 4 score	Score > 2.67 (High risk for advanced fibrosis in liver disease)	High	HCP Alert: Patient [Patient Name] FIB-4 score high. Consider referral. Patient Alert: Your liver test (FIB-4) is elevated. Discuss with your doctor.
27	Vaccinations- Pneumococcal and	60 plus years (Overdue for recommended	Low	HCP Alert: Patient [Patient Name] due

	Influenza	vaccination based on age/condition)		for [Vaccine Type] vaccination. Remind patient/schedule. Patient Alert: It's time for your [Vaccine Type] vaccination. Please talk to your doctor.
	Zoster vaccine	• 50 plus years		
28	eGFR	• <60 mL/min - Reassess medications requirement and dose (Pop up notification) • <30 - Reassess medications requirement and dose (Pop up notification)	High Critical	HCP Alert: Patient [Patient Name] eGFR indicates CKD Stage 3. Review renal protective strategies. Patient Alert: Your kidney function (eGFR) is lower than normal. Please consult your doctor. HCP Alert: Patient [Patient Name] eGFR indicates advanced CKD. Urgent nephrology referral. Patient Alert: Your kidney function is significantly low. Urgent medical review is needed.
29	Weight (Only for patients with heart failure or chronic kidney disease or chronic liver disease)	• Rapid weight gain/loss (e.g., >5% body weight in 1 month, unexplained)	High	HCP Alert: Patient [Patient Name] rapid weight change. Investigate cause. Patient Alert: You've had a significant weight change. Please discuss this with your doctor.
30	Dental symptoms	• Pain or infection		
31	Foot symptoms	• Dryness or itching or any painful cracks in the skin • Ulcer or breach in the skin or between the toes • Frequent wound and infections in feet • Recent problems with		

		fitting your chappals Any change in size and shape of the foot		
32	Changes in family dynamics	- Caregiver health - Death or sickness among any other family member		
33	Follow up visit consultation overdue visit Internal medicine/ Endocrine	Overdue by >7 days (Internal medicine/Endocrine)	High	HCP Alert: Patient [Patient Name] Internal Med/Endo follow-up overdue. Initiate outreach. Patient Alert: Your follow-up visit is overdue. Please schedule it soon.
	- Cardio - Ophthal - Neuro - Nephro	Overdue by >14 days (Cardio, Ophth, Neuro, Nephro)	High	HCP Alert: Patient [Patient Name] specialist follow-up overdue. Expedite appointment. Patient Alert: Your specialist visit is overdue. Please schedule it urgently.
34	Urine routine	Presence of Ketones (trace or more) in diabetic patient with increased blood glucose and especially those on dapagliflozin	High	HCP Alert: Patient [Patient Name] new proteinuria/hematuria. Investigate kidney/urinary issue. Patient Alert: Your urine test shows new findings. Please consult your doctor.
		Significant proteinuria / hematuria (new onset)	High	
35	Symptoms and reports for direct referral to cardio? (Really needed?)	Acute Chest Pain + New or dynamic ECG changes / Troponin elevation	Critical	HCP Alert: Patient [Patient Name] acute cardiac event suspected. Immediate ER Referral. Patient Alert: You have urgent symptoms. Go to the nearest emergency room immediately.

36	Symptoms and reports for direct referral to neuro?	Sudden onset severe headache + focal neurological deficits (e.g., limb weakness, speech disturbance)	Critical	HCP Alert: Patient [Patient Name] acute neurological event suspected. Immediate ER Referral. Patient Alert: You have urgent neurological symptoms. Go to the nearest emergency room immediately.
37	Symptoms and reports for direct referral to emergency?	Unresponsive, severe shortness of breath, acute severe pain, new focal neurological deficit, significant bleeding, severe hypoglycemia (<50 mg/dL)	Critical	HCP Alert: Patient [Patient Name] requires immediate emergency care. Urgent ER Referral. Patient Alert: You have critical symptoms. Call emergency services or go to ER immediately.
38	Symptoms and reports for direct referral to nephro?	Rapidly rising creatinine (e.g., >50% within 48 hrs), anuria, severe hyperkalemia	Critical	HCP Alert: Patient [Patient Name] acute renal failure suspected. Urgent nephrology referral. Patient Alert: You have urgent kidney symptoms. Seek immediate medical attention.
39	Symptoms and reports for direct referral to podiatrist?	New foot ulcer, severe foot pain with fever, signs of infection (redness, warmth, pus) in foot	High	HCP Alert: Patient [Patient Name] new foot pathology. Urgent podiatry/surgical referral. Patient Alert: You have new foot symptoms. Please consult a podiatrist or your doctor urgently.
40	Symptoms and reports for direct referral to surgeon?	Acute abdomen (severe pain, rigidity), severe infection requiring drainage, cellulitis presenting with fever and leg swelling, foot ulcer	Critical	HCP Alert: Patient [Patient Name] requires surgical evaluation. Urgent surgical consult/ER. Patient Alert: You have urgent

				symptoms. Seek immediate emergency medical help.
41	Integrated prescription	• New Medication prescribed by another department (without Primary HCP Review/Acknowledgement): A new medication is added to the patient's record by a specialist (e.g., Cardiology, Nephrology) and has not been acknowledged or reviewed by the patient's primary HCP within [e.g., 24-48 hours]. • Therapeutic Duplication Detected: Two or more medications prescribed by different departments are for the same therapeutic class or indication, potentially leading to overdose or redundancy (e.g., two different statins, two different ACE inhibitors) • Missing Key Prescription Elements: A recently added prescription is missing critical information (e.g., clear dosage, frequency, route, duration, or prescribing physician details) that deviates from standard prescribing templates. • Medication Contradiction for Current Diagnosis/Lab: A	Moderate High High High	HCP Alert: Patient [Patient Name] has a new prescription ([Medication Name]) from [Specialist Department]. Please review for overall care plan alignment and potential interactions. Patient Alert: (Not directly applicable, HCP-focused notification) HCP Alert: Patient [Patient Name] has a potential therapeutic duplication with [Medication A] and [Medication B]. Review with prescribing departments. Patient Alert: (Not directly applicable, HCP-focused) HCP Alert: Patient [Patient Name] new prescription for [Medication Name] has missing details. Review and complete information. Patient Alert: (Not directly applicable, HCP-focused for data integrity) HCP Alert: Patient [Patient Name] [Medication Name] may be contraindicated

		prescribed medication is generally contraindicated based on the patient's <i>known and recent</i> diagnosis or a <i>critical</i> lab result (e.g., Metformin for eGFR <30, ACEI for severe hyperkalemia). (This can be rule-based for <i>known</i> contraindications, though complex interactions are ML).		based on recent [Diagnosis/Lab Result]. Urgent review. Patient Alert: (Not directly applicable, HCP-focused for safety)
	Preferred medication recommendations, Pop up notification	• Patients with MASLD/ HF (Heart failure)/ Renal dysfunction/ Diabetic nephropathy with obesity- GLP 1 agonist- Semaglutide/ Liraglutide • Dual peptide agonist- Tirzepatide • Low GFR, HF patients- Dapagliflozin/ Empagliflozin • To add or titrate ACEi/ARBs- Ramipril, Telmisartan, Olmesartan • To add or titrate Antiplatelet drugs- Aspirin, Clopidogrel • To add or titrate Statin- Atorvastatin, Rosuvastatin • To add or titrate for HFrEF- Beta blockers, ARNI, Mineralocorticoid receptor antagonist (MRA)- Spironolactone/ Aldactone/ Eplerenone/Finerenone, SGLT2i- Dapagliflozin/ Empagliflozin - Reasoning for not putting up these drugs, if it's not prescribed • To add or titrate for		

		HFpEF- MRA, SGLT2i		
42	Habit	Consistent reporting of smoking, excessive alcohol without improvement plan	Moderate	HCP Alert: Patient [Patient Name] persistent unhealthy habits. Recommend behavioural/psychiatry counseling. Close monitoring with trackers and strategies Patient Alert: Let's work on improving your daily habits for better health.
43	Education prescription structure	Patient has not completed/accessed recommended educational modules for >7 days	Low	HCP Alert: Patient [Patient Name] not engaging with education. Remind/re-evaluate needs. Patient Alert: We've sent you new educational material. Please review it.
44	Missing data uploads	- Patient has not uploaded any data (BP, BG, etc.) for >24 hours (or as per defined schedule) - Physician advised - Routine	High/ critical Low	HCP Alert: Patient [Patient Name] missing data uploads. Follow up with patient. Patient Alert: Please remember to upload your health data regularly.
45	Overdue tests	- Recommended lab tests (e.g., Kidney Function, Lipid Profile, Urine ACR) overdue by >1 month - Physician advised - Routine	High/ critical Low	HCP Alert: Patient [Patient Name] has overdue lab tests. Remind/schedule. Patient Alert: You have some important lab tests that are due. Please schedule them.
46	Timelines for data/report uploads	Patient consistently uploads data/reports late by >X hours/days (configurable threshold)	Low	HCP Alert: Patient [Patient Name] consistently late with data uploads. Counsel on consistency. Patient Alert: Consistency in data uploads helps

				your doctor monitor your health better.
47	Data completeness	• Critical data fields missing from patient profile (e.g., baseline weight, height, comorbidities)	Moderate or High	HCP Alert: Patient [Patient Name] profile incomplete. Request missing data. Patient Alert: Your health profile is incomplete. Please provide missing information.
48	Checklist completion for readiness for consultation	• Patient-side: Patient-assigned pre-consultation tasks (e.g., latest vital readings uploads, symptom questionnaires, pre-read educational materials) are not completed by 12 hours prior to the scheduled consultation. • HCP-side: Critical pre-consultation data review (e.g., latest lab results, recent critical alerts, overall patient trends from the platform) is not marked as completed by the HCP by 4 hours prior to the scheduled consultation.	Moderate?	HCP Alert: Patient [Patient Name]'s pre-consultation checklist or your pre-consultation review is pending for the upcoming appointment on [Date/Time]. Please complete/review to ensure an effective consultation. Patient Alert: Your pre-consultation checklist for your upcoming visit on [Date/Time] is incomplete. Please complete it to help your doctor prepare for your visit and make the most of your time.

Alert levels

- **Critical (Red):** Requires immediate attention (within minutes to a few hours) and direct HCP intervention, potentially involving emergency services or urgent referral. These are life-threatening or rapidly progressing conditions.
- **High (Orange/Yellow):** Requires attention within 24-48 hours. Indicates a significant deviation from target or a worsening trend that needs prompt clinical review and potential adjustment to the care plan.
- **Moderate (Blue/Yellow):** Requires attention within 1-3 days. Indicates a noteworthy trend or deviation that should be addressed during the next routine interaction or a focused follow-up.

- **Low (Green/Grey):** Informational or gentle reminder. No immediate clinical danger but highlights areas for improvement in adherence, data logging, or preventative care.

This rule-based engine provides a robust framework. As the platform matures, these rules can be refined and augmented by the advanced predictive analytics from the ML models to create a truly intelligent and responsive system.